

E-Commerce Customer & Purchase Analytics Dashboard

Stage 4 - Documentation, Insights and Presentation

This project dashboard combines customer behavior analysis, engagement trends, funnel performance, device usage patterns, and purchase-related insights into one interactive Power BI report. Python was used for data preparation and analysis, and the results were visualized in Power BI with a clean, aligned dashboard layout.

Project Title	E-Commerce Customer & Purchase Analytics
Tools Used	Python, Pandas, NumPy, Matplotlib, Seaborn, Excel, Power BI
Main Goal	Understand customer behavior and improve purchase conversion
DashBoard Focus	Engagement, funnel stages, device usage, ads, discounts, retention

What the Dashboard Shows

- The dashboard provides a complete overview of customer behavior, engagement, and purchase trends in an e-commerce platform
- It displays important business KPIs such as Total Customers, Purchase Rate, Engagement Score, Returning Customers, and Average Time on Site
- The dashboard visualizes customer demographics through Gender Distribution and Age Group analysis
- The Customer Journey Funnel shows how users move through different engagement stages and identifies conversion flow
- Device Usage analysis highlights customer preferences for Desktop, Mobile, and Tablet devices across different age groups
- The Purchase Rate by Ads & Discounts matrix explains how advertisements and discounts influence customer purchasing behavior
- The Customer Performance Analysis table compares customer segments using metrics such as engagement score, bounce rate, and purchase rate

- Interactive slicers allow users to dynamically filter and explore the data for deeper analysis
- Overall, the dashboard helps businesses understand customer interactions, improve engagement strategies, optimize marketing campaigns, and increase purchase conversions

Final Dashboard

The dashboard below is the final Power BI output, with visuals aligned on a single canvas for a clean presentation.



Findings, Insights, and Business Story

Summary

The dashboard helps understand how customers interact with the e-commerce platform and what factors influence their purchasing behavior. It shows that mobile devices are the most commonly used platform among customers, and highly engaged users are more likely to make purchases.

The analysis also reveals that advertisements and discounts positively affect customer buying decisions. Returning customers contribute significantly to overall engagement, while low-retention users tend to leave the platform quickly, resulting in higher bounce rates.

By analyzing customer demographics, engagement levels, and device preferences, the dashboard provides useful insights that can help businesses improve customer experience, increase engagement, and optimize sales and marketing strategies.

Key insights

- Mobile users appear to form a major part of the audience, so mobile experience matters a lot.
- Higher engagement is generally associated with better purchase outcomes and lower bounce behavior.
- Discounts and ad clicks help reveal which marketing actions are more likely to influence conversion.
- Returning customers contribute significantly to overall platform activity, indicating strong customer retention value
- Low retention users exhibit high bounce rates, highlighting the need for personalized engagement and retention strategies

Recommendations

- Improve the mobile user experience by optimizing website performance and responsiveness for mobile devices
- Use personalized discounts and targeted advertisements to increase customer engagement and purchase conversions
- Focus on retaining highly engaged customers through loyalty programs and personalized recommendations
- Implement re-engagement strategies for low-retention users to reduce bounce rates and improve customer retention
- Analyze customer behavior regularly to identify changing trends and optimize marketing strategies accordingly
- Enhance product recommendation systems to improve customer satisfaction and increase sales opportunities

Final Story with the Dashboard

The dashboard transforms raw customer data into meaningful business insights through interactive visualizations and KPI analysis. It explains how customer engagement, device usage, demographic factors, and promotional strategies influence purchasing behavior.

The dashboard enables businesses to monitor customer activity, identify high-value customer segments, and improve conversion strategies. By combining multiple analytical views in a single platform, it supports effective decision-making and helps enhance customer experience and business growth.